SPACES

iMPACTPack

Making Biodiversity Learning Matter

by Actually Increasing Local Biodiversity and Proving It Stayed Up

Students take action to increase biodiversity in a specific local area - planting, creating habitats, removing invasives - and prove with survey data that species count went up and stayed up.

Domain	Evidence Window	Grading	AI Integrity
Spaces	12 to 16 weeks	Delayed (likely end of next unit)	High: field survey data with dated photos at a real location

The Opportunity

Every school teaches ecosystems and biodiversity. With iMPACT, that learning becomes an opportunity for students to genuinely increase the diversity of life in a real local space. Over 12 to 16 weeks, students conduct baseline surveys, plan and implement habitat interventions, and collect follow-up survey data to prove that species count increased and stayed up.

This is an opportunity to make local ecosystems better through learning that leaves living proof.

The iMPACT Challenge

Your Opportunity

Increase biodiversity in a specific local area and sustain that increase for at least 12 weeks. Your evidence is your survey data. Your grade reflects the positive change you made to the living world.

What Students Do

After studying ecosystems in class, students take their learning outside and apply it. The teaching unit finishes. The iMPACT challenge continues in the background as normal school life goes on. Students:

- **Survey** their chosen area to establish a baseline species count using a standardised method
- **Plan** specific, realistic interventions: planting native species, creating microhabitats, removing invasive plants, installing wildlife features
- **Implement** those interventions and document the process
- **Conduct** follow-up species surveys at monthly intervals using the same method
- **Complete** a final survey showing sustained increase compared to baseline
- **Reflect** on ecosystems, intervention effectiveness, and what biodiversity means locally

Accepted Evidence

- Baseline species survey using a standardised method (quadrat, transect, point count, pitfall trap)
- Dated photographs of the site before, during, and after interventions
- Intervention plan with rationale linked to ecology
- Follow-up survey data at regular intervals (minimum monthly)
- Final survey data showing sustained species increase
- GPS coordinates or clear site identification

About AI Integrity

The iMPACT model places the student in the centre of real-world ecological change. Field survey data collected at a specific location with dated photographs, GPS tagging, and repeated site visits creates a chain of evidence rooted in physical reality. The effort to fabricate months of consistent, plausible ecological survey data at a real location far exceeds the effort to simply do the work.

Grading Timeline

The teaching unit ends. School life continues. The iMPACT evidence builds in the background over 12 to 16 weeks. The grade for this challenge will likely arrive end of the following unit. This is by design: ecosystems respond slowly, and real ecological impact takes time.

Students who increase biodiversity in their local environment carry that impact - literally, in living organisms - long after the grade is recorded.

Evidence Guide

Students build a portfolio of real-world proof over the iMPACT window. This portfolio is the assessment submission. It grows in the background as students live the challenge.

Evidence Item	What It Shows	Format
Baseline Survey	Species count and diversity at the chosen site before intervention: the starting point	Survey data sheet, photos, method description
Intervention Plan	Specific actions the student committed to, with ecological rationale (max 1 page)	Written document
Implementation Log	Dated photos and notes documenting what was done, when, and where	Digital log or journal
Monthly Surveys	Species surveys at regular intervals using the same method as baseline	Survey data sheets, photos
Final Survey	End-of-window survey showing sustained increase compared to baseline	Survey data sheet, photos
Reflection	What worked, what surprised them, what biodiversity means locally and why it matters (max 500 words)	Written document

Timeline

Phase	What Happens
Teaching	Ecosystems unit taught as normal. Final sessions introduce the iMPACT Challenge. Students identify potential survey sites and learn survey methods.
Handover	Teaching moves on. Students conduct baseline surveys, write intervention plans, begin implementing changes. Parent letter goes home.
iMPACT Window	Students live the challenge. Monthly surveys maintained. Brief class check-ins every 3 to 4 weeks. Other teaching continues.
Portfolio Submission	Evidence portfolios submitted. Will likely fall at the end of the following unit. Students share data and site stories.
Assessment	Impact graded. Results may be reported next cycle. Collective class biodiversity gains celebrated.

The delayed timeline is a feature. Ecosystems respond slowly. Schools should plan this unit early enough to allow the full evidence window, ideally spanning a seasonal transition.

Rubric: Band Descriptors

The iMPACT rubric assesses two dimensions: the real-world change achieved, and the quality of evidence and reflection supporting it. The top bands are reserved for students whose impact is visible, sustained, and clearly connected to their learning.

Band	Grade	Positive Impact Achieved	Evidence and Reflection
Deep Impact	**A / 7 to 8**	Significant, sustained increase in species diversity over 12+ weeks, clearly attributable to student interventions. Evidence of ecological thinking in intervention design. Site shows visible habitat improvement.	Complete, authentic portfolio with rich survey data across multiple time points. Sophisticated connections between ecology and intervention decisions. Deep understanding of why certain actions worked and what biodiversity means in context.
Meaningful Impact	**B / 5 to 6**	Clear, sustained increase in species count over 12+ weeks. Interventions directly linked to student actions. Improvement maintained across final surveys.	Complete portfolio with clear survey data showing upward trend. Reflection connects classroom learning to field outcomes. Student articulates what worked and what was challenging.
Growing Impact	**C / 3 to 4**	Some increase evident but may be inconsistent or partially sustained. Genuine effort with interventions implemented, but ecological response is modest or uneven.	Portfolio mostly complete, may have gaps in survey intervals. Reflection shows some connection between ecology learning and fieldwork.
Emerging Impact	**D / 1 to 2**	Minimal measurable change. Student engaged but implementation was limited or surveys were too infrequent to demonstrate sustained impact.	Portfolio incomplete or inconsistent. Reflection brief or does not connect meaningfully to ecosystems learning.
No Impact	**N/A**	No evidence of sustained effort toward the challenge.	Portfolio not submitted or no authentic evidence.

The impact is the grade. A beautifully written reflection with no real survey data cannot score above Growing Impact. Authentic field data showing genuine, sustained biodiversity increase, even with rough presentation, can reach Meaningful or Deep Impact.

Contextual Fairness

Students work in vastly different environments. A student in an urban concrete area who attracts three new insect species to a planter box may demonstrate more ecological ingenuity than a student in a lush garden who plants wildflowers. The rubric rewards proportional change relative to starting point, sustained effort, and genuine engagement with the ecology of their specific site.

Teacher Setup Guide

This guide walks you through the iMPACT journey in four phases. Each has clear milestones. The iMPACT window runs in the background while your teaching moves on.

Phase 1: Prepare (1 to 2 weeks before the unit)

- Identify potential survey sites. School grounds, local parks, community gardens, home gardens. Get permissions where needed.
- Establish baseline protocols. Decide which survey methods are appropriate and feasible: quadrats, transects, point counts, pitfall traps. Students should use methods they can repeat independently.
- Send the parent/guardian letter home. Allow two weeks for responses.
- Source basic survey equipment if budget allows. Quadrat frames, hand lenses, field guides, identification apps (iNaturalist, Seek). Useful but smartphones alone can work.
- Plan the seasonal window. Biodiversity surveys are more meaningful if they span a seasonal transition (e.g. dry to wet, winter to spring).

Phase 2: Teach (Your normal unit)

- Teach the ecosystems unit as you normally would. iMPACT changes the assessment endpoint, not your teaching.
- In the final 1 to 2 lessons, introduce the iMPACT Challenge. Share the challenge, walk through evidence guide and rubric.
- Students identify their survey site and conduct baseline surveys in class or as a supervised field trip.
- Students draft their intervention plan. Max 1 page. Encourage ecological thinking: what does this site need? What species are missing and why?

Phase 3: Live It

- Move on with your teaching. The challenge runs in the background.
- Run brief check-ins every 3 to 4 weeks (5 to 10 minutes). Ask one or two students to share a survey update or site photo.
- Monitor the shared tracking space. Glance periodically. A quick private word for students who haven't surveyed in a month.
- Celebrate early sightings. New species appearing? A bird using a nesting box? Acknowledge it.
- Expect and welcome variability. Weather, seasons, animal behaviour: these are real ecology. Not every survey will show an increase.

Phase 4: Harvest (circa 12 weeks later)

- Give a clear submission deadline 2 weeks in advance.
- Make submission day a celebration. Species gallery wall, site photo presentations, or a class biodiversity map.
- Grade using the rubric. Focus on survey data trends first, then evidence quality and reflection.
- Calculate the class's collective impact. Total new species documented. Total area improved. Powerful for assemblies and reporting.
- Share results with the local community. If students worked in public spaces, share outcomes with parks departments, community groups, or local media.

Frequently Asked Questions

What if species count doesn't increase despite genuine effort? Ecosystems are complex and don't always respond on our timeline. A student demonstrating genuine sustained effort with honest data, thoughtful analysis of why change didn't occur, and strong ecological reflection can still achieve a Growing Impact grade.

What about students in urban environments with limited green space? Urban biodiversity is real and valuable. Container gardens, balcony planters, window boxes, or even a single tree can be surveyed. Insect hotels, bee-friendly planting, and bird feeders in small spaces are legitimate interventions. The rubric rewards proportional change.

Can students work in pairs or groups on the same site? Students may share a site and collaborate on interventions. Each submits their own baseline data, survey logs, and reflection. Shared implementation photos are acceptable if each student's individual contribution is clear.

What if a student can't access an outdoor space regularly? Adapted challenge: monitoring and improving biodiversity in a school garden or specific school ground area, with regular access during school hours. Teacher oversight replaces independent site visits.

What survey method should students use? Match the method to the site and the organisms being tracked. Quadrats for plants and ground invertebrates, point counts for birds, pitfall traps for ground beetles (with teacher guidance on ethics and safety). Consistency matters more than complexity, the same method at the same site each time.

When does the grade get reported? Likely the following semester. The 12 to 16 week evidence window means this challenge will almost certainly cross a reporting boundary. Agree with your assessment coordinator before the challenge begins.

Parent/Guardian Letter Template

Adapt to your school context. Students working on spaces at home will need household permission and support.

Dear Parent/Guardian,

As part of our commitment to making learning matter, your child is taking part in an iMPACT Challenge: an opportunity to make a real, positive difference to local biodiversity by increasing species diversity in a chosen area through what they are learning in [Science/Geography].

To support this, your child will need:

- Access to an outdoor area they can visit regularly to survey and improve (this could be a home garden, local park, school grounds, or community space)
- Your support for regular site visits over the next 12 to 16 weeks
- A willingness to allow small habitat improvements if the site is at home (planting, nesting boxes, etc.)

Your child's assessment will be based on the real, positive change they achieve and sustain. We believe that when learning leads to real impact, it matters more and lasts longer.

You may see iMPACT challenges in different subjects throughout your child's time at [school name]. Each one is an opportunity for your family to make the world a little better through learning. Thank you for being part of it. If your situation makes participation difficult, please get in touch and we will arrange an alternative.

[Teacher name] | [School name]

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